Customer Segmentation Using K-Means and Hierarchical Clustering

An Unsupervised Machine Learning Approach to Customer Behaviour Analysis

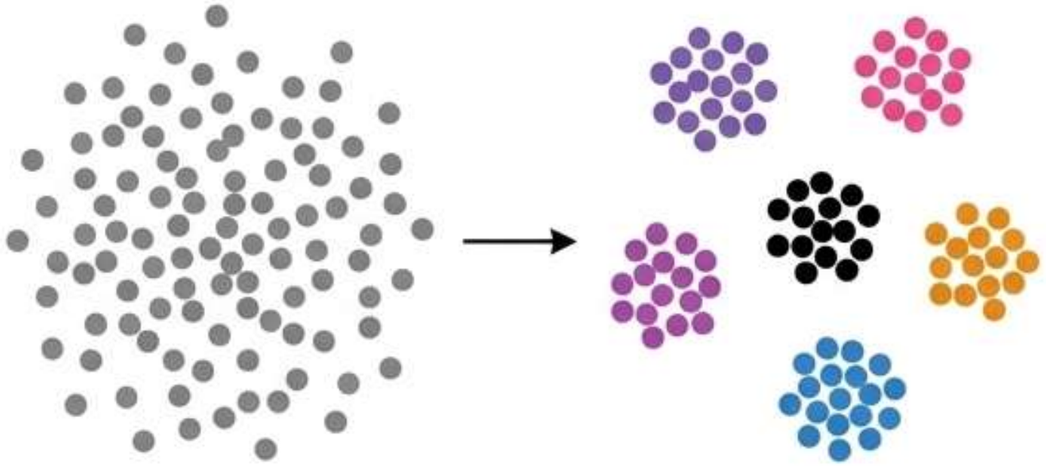
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Executive Abstract

Customer segmentation is a crucial marketing strategy that divides a market into distinct groups based on shared characteristics. This project presents an end-to-end unsupervised machine learning pipeline for customer behaviour analysis using a synthetic dataset of 1,000,000 records. We implement and evaluate two distinct clustering algorithms: K-Means Clustering and Agglomerative Hierarchical Clustring. The models utilize core features such as Annual Income and Spending Score to mathematically identify distinct customer personas. To handle the computational complexity of the massive dataset, feature scaling and strategic sampling are employed. The clustering quality is rigorously evaluated using the Silhouette Score and the Elbow Method. Finally, the optimized K-Means model is deployed via an interactive Gradio web dashboard, bridging academic rigor with practical business application. The system provides real-time segment prediction and actionable business intelligence, demonstrating the powerful intersection of data science and marketing strategy.

The Business Imperative of Customer Segmentation



Definition: The process of dividing a target market into distinct groups based on shared characteristics, behaviours, or preferences.

Targeted Personalization
Moves marketing beyond a one-size-fits-all approach to tailored messaging.

Resource Optimization
Focuses marketing spend on high-value customer segments, improving overall ROI.

Customer Loyalty
Anticipates specific segment needs, fostering long-term brand relationships.

Problem Statement and Project Objectives

Given a massive customer database, how can a business mathematically identify underlying spending patterns without pre-existing labels, and make these insights easily accessible to non-technical stakeholders?



1. Analyze & Process

Explore a massive 1-million record dataset, focusing on Annual Income and Spending Score.



2. Cluster & Compare

Apply and contrast K-Means and Agglomerative Hierarchical Clustering, utilizing evaluation metrics like the Silhouette Score.



3. Deploy & Act

Develop an interactive Gradio web application that translates mathematical clusters into real-time business strategies.

The Raw Material: Customer Dataset

Dataset Overview

1,000,000

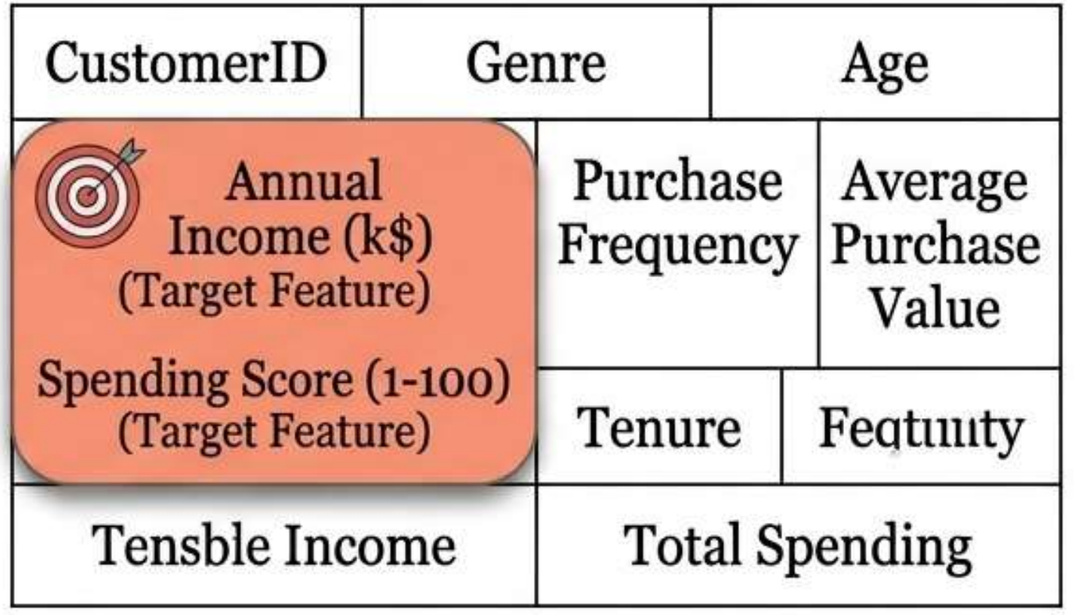
Total Records

9

Total Columns

Nature: Synthetic dataset explicitly designed for large-scale ML clustering demonstrations.

Available Features



Methodological Choice: We isolated Income and Spending Score to create clear, visually interpretable 2D business segments.

Tools and Technologies Stack

Environment & Version Control

- Google Colab: Cloud execution
- GitHub & GitHub Pages: Version control and project documentation

Data Processing & Visualization



- Pandas & NumPy: High-performance data manipulation on 1M records.
- Matplotlib & Seaborn: Statistical visualization and feature distribution plotting.

Machine Learning Engines



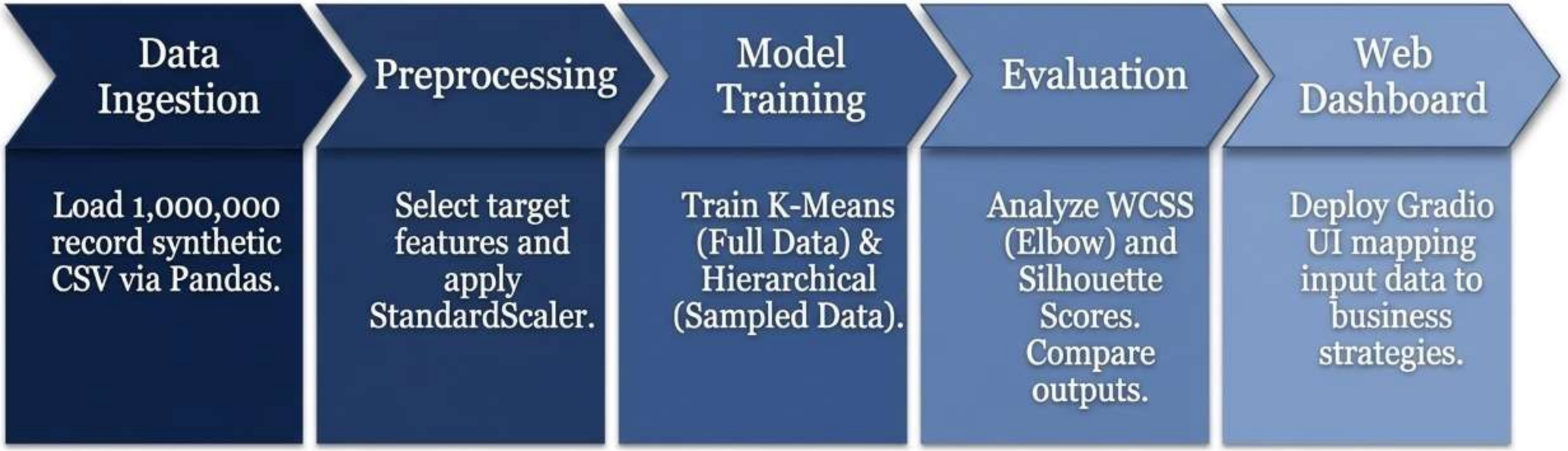
- Scikit-learn (sklearn): K-Means clustering, preprocessing, and evaluation metrics.
- SciPy: Agglomerative hierarchical linkages and dendrogram generation.

Web Deployment



- Gradio (Python) / HTML / CSS / JavaScript: Underpins the interactive web dashboard for real-time inference.

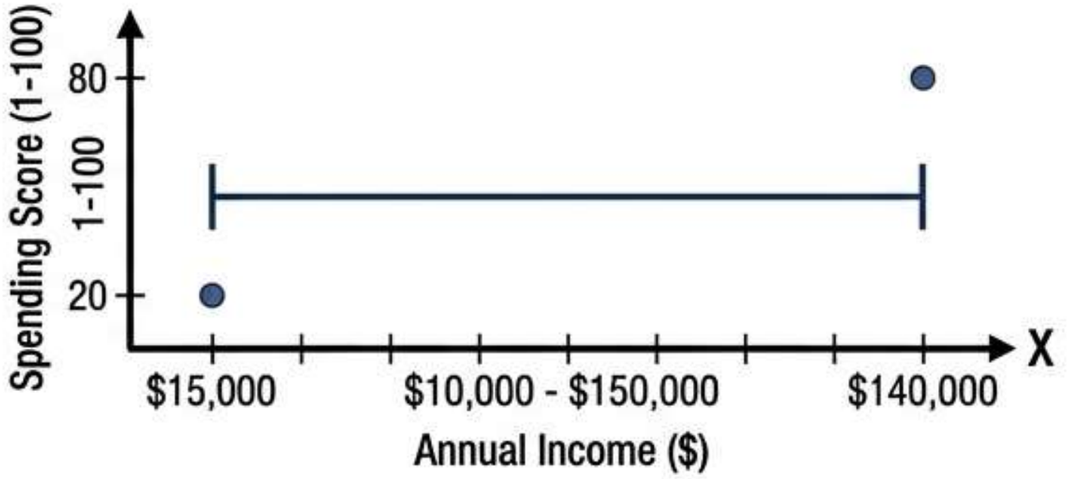
System Architecture and Workflow



Data Preprocessing: The Mathematics of Scaling

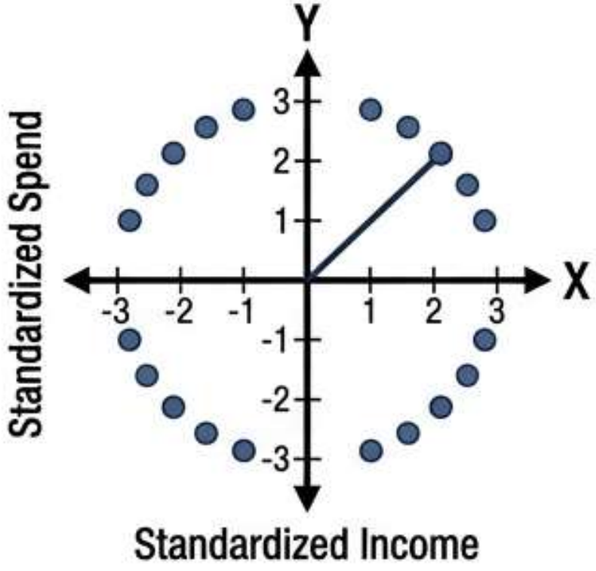
Feature Selection: Extracted Annual Income and Spending Score as a 2D matrix.

The Problem with Raw Data



- Clustering algorithms (K-Means, Hierarchical) rely on Euclidean Distance metrics.
- Without scaling, the algorithm would incorrectly treat Income as the only mathematically significant variable.

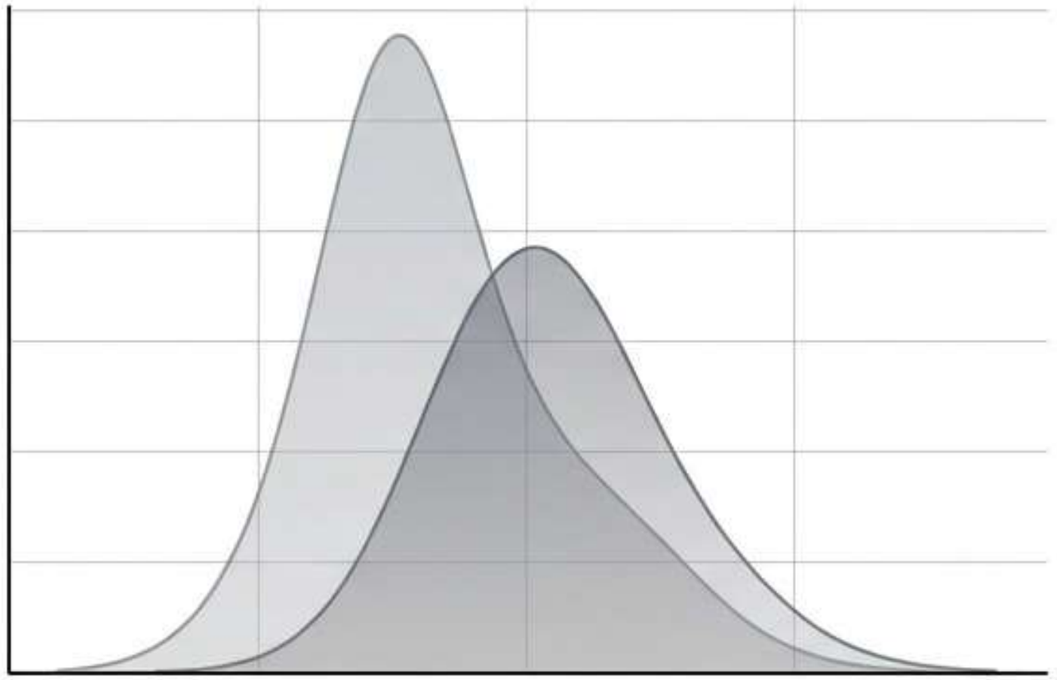
The Solution: StandardScaler



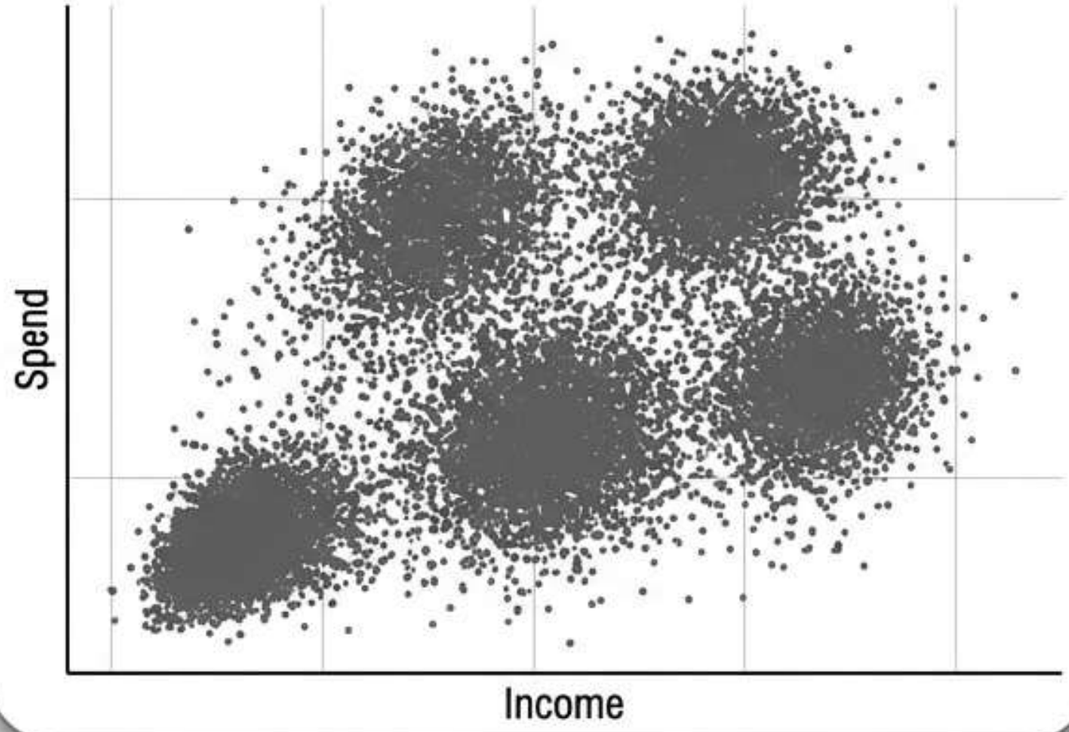
- Data is standardized to have a mean of 0 and a variance of 1.
- Ensures both dimensions contribute equally to the spatial distance between data points.

Exploratory Data Analysis (EDA)

Feature Distribution



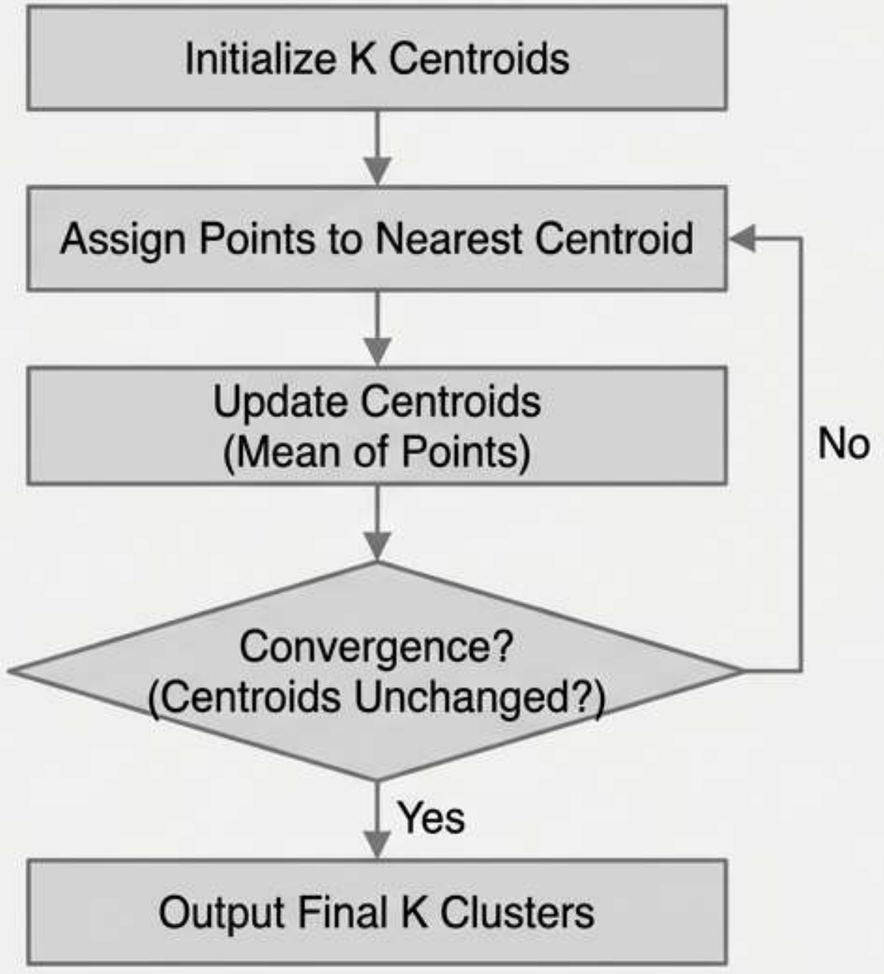
Pre-Clustering Scatter Plot



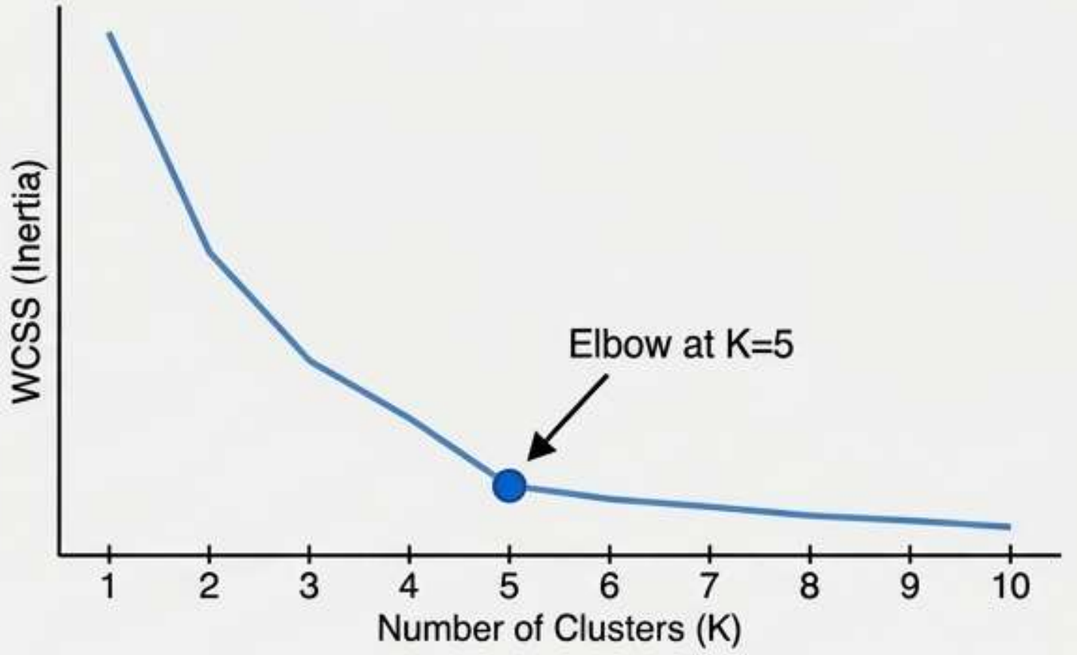
Observed Patterns: The 2D scatter plot reveals natural densities and emerging groupings before any algorithm is applied. These visual densities confirm that distance-based clustering algorithms will be effective in this multi-dimensional space.

K-Means Clustering & The Elbow Method

Algorithm Mechanics



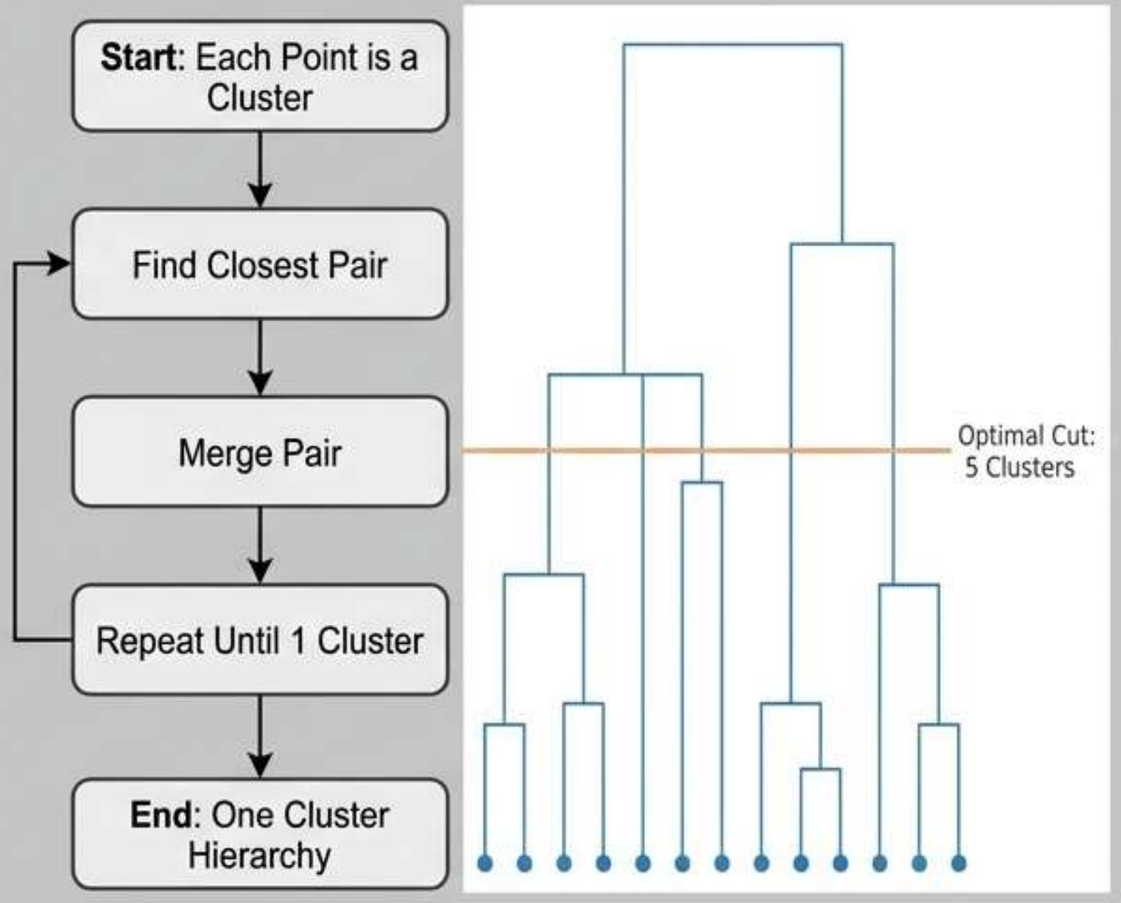
Finding Optimal K



- We calculate the Within-Cluster Sum of Squares (WCSS).
- The elbow indicates where adding clusters yields diminishing returns. Based on our analysis, K=5 was chosen as the mathematically optimal cluster count.

Agglomerative Hierarchical Clustering

Algorithm & Dendrogram

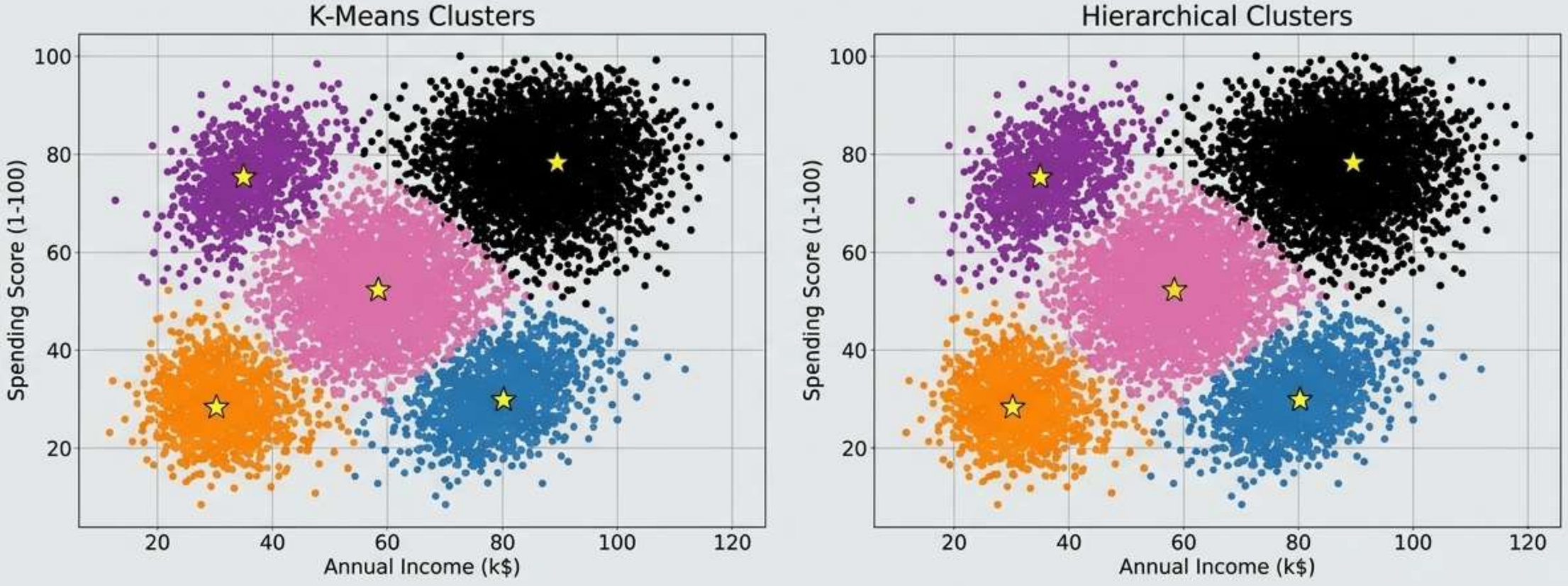


Technical Constraints & Analysis

- **Methodology:**
  - **Bottom-Up Approach**: Starts with every point as an individual cluster and iteratively merges the most similar pairs.
  - **Linkage Criterion**: Utilized Ward Linkage (minimizes the variance of clusters being merged).
- **Visual Analysis**: The longest vertical lines without horizontal intersection confirm our choice of 5 clusters.
- **Computational Reality**:
  - Hierarchical clustering has a memory complexity of $O(N^2)$.
  - To process the 1,000,000 record dataset without memory exhaustion, we successfully applied the algorithm to a representative sample using SciPy.

Results: 5 Distinct Customer Personas

Comparing wo plots of the final segmented clusters



- Target (Target Focus): High Income, High Spend (Upper Right)
- Careless (Impulsive): Low Income, High Spend (Upper Left)
- Sensible (Frugal): Low Income, Low Spend (Lower Left)
- Standard (Average): Medium Income, Medium Spend (Center)
- Careful (Savers): High Income, Low Spend (Lower Right)

Model Evaluation & Algorithm Comparison

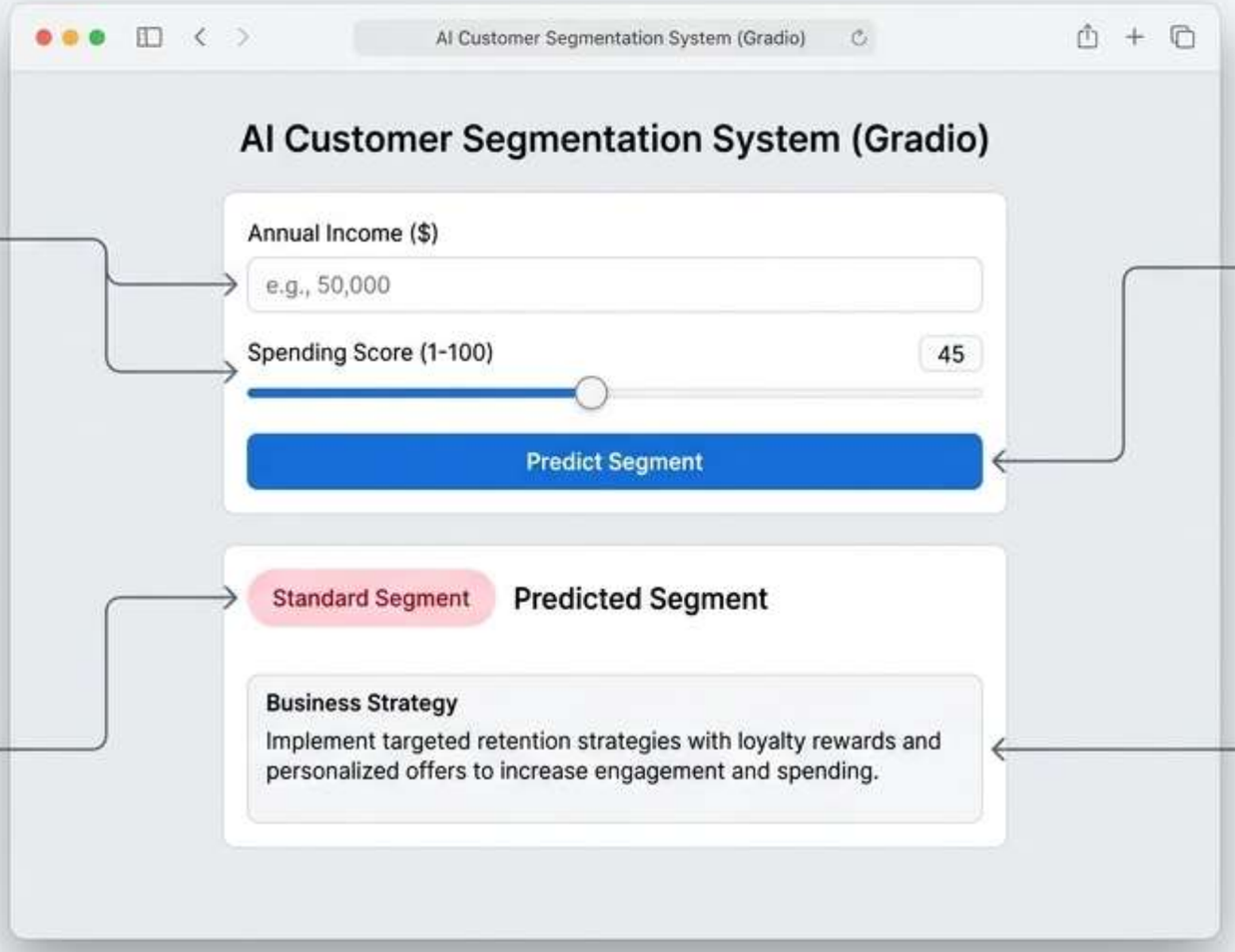
Dimension	K-Means Clustering	Agglomerative Hierarchical
Methodology	Centroid-based (Top-down)	Linkage-based (Bottom-up)
Scalability (1M Records)	Highly efficient, processes full dataset rapidly.	Memory heavy $O(N^2)$, required data sampling.
Predefined Clusters (K)	Requires predefined K (Found via Elbow Method).	Does not require predefined K (Found via Dendrogram).
Evaluation Focus	Evaluated via Silhouette Score measuring internal cohesion vs external separation.	Slower execution but provides granular hierarchical relationships.

Conclusion: K-Means was selected as the superior engine for our live dashboard due to its unmatched efficiency on the full 1-million record dataset.

Live Deployment: AI Customer Segmentation System

1. Feature Input

Users input an Annual Income and adjust a Spending Score slider.



2. Inference Engine

The backend instantly calculates the scaled Euclidean distance to the pre-trained K-Means centroids.

3. Output Result

The system predicts the exact Customer Segment.

4. Business Strategy

Generates a tailored marketing recommendation based on the predicted persona.

Test Cases & Scenario Validation

Diagnostic table

Annual Income	Spending Score	Model Prediction	Validation Status
$86,000	80	Target (High Income / High Spend)	⏳ (Awaiting Verification)
$25,000	20	Sensible (Low Income / / Low Spend)	⏳ (Awaiting Verification)
$55,000	50	Standard (Mid Income / Mid Spend)	⏳ (Awaiting Verification)

Note: Predicted clusters are mapped to the logical quadrants of our trained 5-cluster model. Marked 'Awaiting Verification' pending final live-server confirmation.

Conclusion and Future Scope

Verified Findings

- Successfully implemented a highly scalable K-Means model on 1M records.
- Identified 5 distinct, actionable customer personas verified by both algorithms.

Limitations

- The 1-million record dataset is synthetic; real-world behavior may be more erratic.
- Hierarchical clustering required sampling due to extreme memory complexity.

Future Scope

- Apply density-based algorithms like DBSCAN for irregular cluster shapes.
- Integrate real-world, licensed transaction data to expand the feature space.

Thank You — Questions?